

FATHIMMA RAFEEK

AI/ML Engineer | Deployment & Orchestration Specialist | Python, PyTorch, LangChain

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PROFESSIONAL SUMMARY

Highly results-driven **AI/ML Engineer** with **1+ years of post-academic experience** in the end-to-end development, **deployment, and maintenance of scalable AI solutions**. Expert in core machine learning, **Deep Learning, Computer Vision (CV), and NLP/LLM orchestration**, utilizing **Python, TensorFlow, and PyTorch**. Proficient in the entire data lifecycle, from **data preprocessing** and pipeline creation to model monitoring. Directly align with the Inception AI Associate requirements through proven experience using **Docker** and cloud services to deliver high-quality, **low-latency, and regulatory-compliant solutions** in production environments. Passionate about contributing to cutting-edge, domain-specific AI orchestration.

PROFESSIONAL EXPERIENCE

AI Developer - Zerosnap ID Solutions | Aug 2025 - Present

- **Pioneered** high-volume **invoice processing** workflows by integrating **Azure Document Intelligence** for superior data extraction accuracy and deploying the resulting logic as **AWS Lambda** functions for efficient, serverless execution.
- **Integrated** advanced computer vision models, including **YOLO-based parsers**, for accurate extraction of structured data from crucial documents like **Passport, Aadhar, and PAN Cards**, enabling fast and compliant KYC workflows.
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- **Collaborated with engineering teams** to **orchestrate model deployment** using **Docker and cloud services**, ensuring **scalability, low latency**, and regulatory adherence across **production systems**.

Key Skills Applied: Model Deployment (AWS Lambda), Azure Document Intelligence, YOLO, Computer Vision, Scalability, Collaboration.

INTERNSHIP EXPERIENCE & FOUNDATION (Data Science & AI) AI

Developer Intern - Jezt AI | Mar 2025 - Aug 2025 (6 Months)

- Built modular **AI applications** integrating **Computer Vision (CV)** and **LLMs**, establishing foundational experience with complex, multi-modal **AI solutions**.

- Developed effective **prompt engineering strategies** and **modular AI workflows**, demonstrating familiarity with modern **Agentic AI** concepts.
- **Spearheaded** the development and **deployment** of high-accuracy **Facial Recognition** and **Liveness Detection models**, significantly increasing identity **verification accuracy** and reducing fraud risk in **real-time user onboarding**.
- **Integrated OCR** and **document classification models** to **extract structured data** from global ID documents, enabling **fast and compliant KYC workflows**.
- **Collaborated with engineering teams** to **orchestrate model deployment** using **Docker and cloud services**, ensuring **scalability, low latency**, and regulatory adherence across **production systems**.
- *Key Skills Applied: Model Deployment, Computer Vision, Scalability, Collaboration.*

Data Scientist Intern - Wahy Lab Solutions | Jan 2025 - Apr 2025 (3 Months)

- Designed and **delivered AI/ML/DL/NLP solutions** for business use cases, focusing on seamless integration into existing operational workflows.
- Contributed to the **training, evaluation, and optimization** of models, resulting in measurable accuracy improvements in **production environments**.

Data Science with Gen AI Intern - Idatalytics | Aug 2024 - Dec 2024 (5 Months)

- **Pioneered** foundational data science workflows, including **Exploratory Data Analysis (EDA)** and advanced **feature engineering** for predictive modeling.
- Developed and deployed **interactive data dashboards** using **Power BI, Tableau, and SQL** directly fulfilling the requirement for experience with data **analysis and visualization tools**.

RELEVANT PROJECTS (Demonstrating Applied R&D and Model Deployment)

AI Agent with Weather Information Retrieval (Focus: Reasoning Agents & Orchestration)

- Developed a conversational **AI agent** using **LangChain** (a key orchestration framework) and HuggingFace for specialized **information retrieval** via the Open Weather API.
- Implemented advanced memory and control flow (using Conversation BufferMemory and **zero-shot-react-description reasoning**), showcasing an understanding of **reasoning agents** and complex conversational structures.
- **Deployed the solution** in an interactive **Streamlit app** for real-time user interaction, demonstrating end-to-end delivery experience.

OCR-Based Certificate Translation Pipeline (Focus: Data Preprocessing & Pipeline)

- Designed an efficient **AI-powered pipeline** to **extract text** from certificate images using **OCR**.
- Applied critical **image pre-processing techniques** (grayscale conversion, noise removal) to significantly enhance **OCR accuracy**—a direct match for the "Collect, clean, and **preprocess data**" responsibility.
- Automated **batch processing** for multiple translations and integrated robust **error handling** for high-volume data reliability.

TECHNICAL SKILLS

Category	Skills
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Programming	Python (Expert), C++, SQL, HTML
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AI/ML	
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	PyTorch, TensorFlow, scikit-learn, LangChain, HuggingFace
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Frameworks	
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	Deep Learning, Computer Vision (CV), NLP/LLMs, Predictive
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Core AI/ML	
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	Modeling, Feature Engineering
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Data & Analysis	
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	Pandas, NumPy, Matplotlib, Seaborn (Data Analysis & Visualization), Power BI, Tableau
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Deployment/Ops	Docker, Cloud Services (Conceptual), Streamlit, Git
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EDUCATION

Master of Computer Applications (MCA) | *Expected 2026* | Indira Gandhi Open University

M.Sc. Physics (Electronics) | *2023* | Mar Athanasius College, Kothamangalam

B.Sc. Physics | *2021* | St. Xavier's College for Women, Aluva

Analytical Foundation: Physics degrees provide a strong mathematical background, rigorous problem-solving training, and systematic scientific inquiry essential for advanced algorithm development.

CERTIFICATIONS

Python Fundamentals | Data Analysis with Python | SQL & Relational Databases

ADDITIONAL ATTRIBUTES

- **Problem-Solving Skills** and **Attention to Detail** are foundational to all project success.
- **Excellent communication and teamwork skills**, proven through collaboration with cross-functional teams and mentorship.
- **Inquisitive mind** and **agility to adapt to ambiguity** demonstrated through rapid deployment cycles in high-stakes identity verification projects.